



Geography Division Transformation and Application Modernization (GEOTAM)

Attachment X: GEOTAM Current Environment

Draft

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1. Purpose

This attachment provides a representative list of the tools and technology currently utilized by GEO. The accuracy of the information may change between the RFQ release and commencement of the call order.

2. Current Environment

2.1. Databases

Environment	#Databases Supported	Example Databases
PRODUCTION (On-Prem, Cloud)	28 – Oracle 1 – PostgreSQL RDS	PRODTRAN, PRODBNCH, REFDB, CQRBNC, PRDGEODB, PRODFLOW
TEST (On-Prem, Cloud)	26 – Oracle 1 – PostgreSQL RDS	TESTTRAN, TESTBNCH, TESTCQR, TSTGEODB, TESTFLOW, TSTCEDSI, TSTCOTS3, TESTDB
DEVELOPMENT (On-Prem, Cloud)	24 – Oracle 1 – Oracle RDS 5 – PostgreSQL RDS	DEVTRAN, DEVBNC, DEVFLOW, DEVGEODB, DEVCEDSI, DEVGIS, IMGDEV

2.2. Tools

The following is a representative list of tools currently utilized by GEO.

Product / Category	Product/Tool
ArcGIS	- Enterprise (Portal for ArcGIS, ArcGIS Server, and ArcGIS Data Store) and extensions - ArcGIS Pro and extensions - ArcGIS Online (AGOL)
ESRI	- ESRI frameworks (deep learning toolkit), SDKs, and API - ESRI Enterprise and File Geodatabases - ESRI Enterprise geodatabase schemas, - ESRI Image Analyst extension, Microfocus Solutions Business Manager (SBM)
Metadata	Saxonica Saxon
Monitoring tools	- Oracle Enterprise Manager (OEM) - AppDynamics - SolarWinds - Sentinel - Splunk Enterprise
Application Server Examples	Tomcat

Product / Category	Product/Tool
Web servers	• Apache • Oracle HTTP Server (OHS)
Business Intelligence tools	• BI Publisher • OBIEE • Oracle Analytics Server
Database Management	• Oracle Database 19c and beyond • Amazon RDS (PostgreSQL, Aurora, Oracle, Redshift)
Low-code Application Development Platform	• Oracle APEX
DevSecOps Tools	• GitLab, • Katalon • Jira • Code Quality • OWASP • Selenium • Jmeter
Container Solution Software	• Amazon Web Services (AWS) • EKS • Podman • Skopeo • Buildah • FluentD • Kibana
Automation tools:	• Ansible • Teraform
GIS tools	• Spatial Studio • Mapviewer
Workflow Management Software	• Redwood Cronacle/RunMyJobs or replacement COTS/Open-Source solution
COTS Mapping Software	• Adobe Experience Manager (AEM) Forms
Cloud Native Tools	

2.3. Enterprise Verification Software

Examples of application software projects to support data quality and enterprise verification include:

- Automated batch quality control software to verify receipt, loading, and aggregation of decennial tabulation population and housing counts.
- Automated processes to verify correct migration and loading of geographic areas.
- Stage Summary Level data to support quality control efforts, e.g., of Geodatabase, Cartographic Boundary Files, and SAIPE File Geodatabase products.

- Maintain, develop, and enhance new software as needed to perform verification and validation of geographic products, including:
 - o Cartographic Boundary Files
 - o Small Area Income and Poverty Estimates (SAIPE) Generalized and Detailed Geodatabases
 - o TIGERweb Geodatabases
 - o TIGERweb 508 HTML Files
 - o Data Product Production (DPP) Tables
 - o Center for Enterprise Dissemination Services and Consumer Innovation (CEDSCI) Geographic Information Tables (GIT)
 - o CEDSCI Geodatabases
 - o CEDSCI Geographic Metadata Repository (GMDR)
 - o Geographic Reference Files (GRF) Codes (GRF-C) and Names (GRF-N)
 - o Master Address File Extracts (MAFX) and associated address products
 - o Product Metadata
 - o Address Count Listing Files
 - o Geographic Update Reporting Logs (GURL)
 - o TIGER/Line Shapefiles
 - o TIGER/Line Geodatabases
 - o TIGER/Line GeoPackage Databases
 - o TIGER/Line GeoPackage Files
 - o Partnership Shapefiles
 - o Live MAFTIGER Partnership Resource Extraction (LMPRE) Shapefiles
 - o Listing and Mapping Application (LiMA) Spatial Extracts
 - o Full Spatial Extracts
 - o QuickFacts
 - o County Adjacency File
 - o Gazetteer Files
 - o Relationship Tables
 - o ZIP Code to County Universe File
 - o New future products
- GoldenGate Support: Performs quality control checks to verify tables and columns containing user-defined data types have the same records and attribute values in an MTDB benchmark created using DataPump export/import and a benchmark created using GoldenGate synchronization.
- Vintage Creation Verification: Performs data comparison quality control checks within the MAFTIGER database to verify the Create Vintage results for each major Benchmark cycle.

- **Vintage Deletion Verification:** Performs data comparison quality control checks with the MTDB to verify the Remove/Delete Vintage process approximately once per year.
- **Data Change Tracking Dashboard:** Provide a user interface to allow users to easily query the audit tables in MAF/TIGER about 'Add', 'Change', and 'Delete' actions.
- **GCAT Update Automated Edit:** Performs a comparison of AREALAND and AREAWATER field values for all geographic features to verify the GCAT Post-Benchmark Update process results. Includes a web interface that allows authorized users to view the automated QC results, determine and post a final, curated QC result, and store relevant documentation about the decision.
- **MAF/TIGER Data Model Monitoring:** Persists data model information about the MAF/TIGER database and performs automated scans to compare MTDB schemas against the data model.
- **Unicode Character Metadata Repository:** Includes automated batch application that discovers and reports non-ASCII Unicode characters. Persists one or more sets of Unicode characters that are approved for selected tables and fields in MAF/TIGER and an automated batch application that discovers and reports any non-approved Unicode characters. Includes a library of PL/SQL Unicode utility functions for use by other applications (e.g., MAF/TIGER Feature Name Standardizer, CoreAPI library folder naming scripts). Includes a web interface for reviewing the results of the scans, and for viewing and updating the approved Unicode character sets.
- Automated checks of Address Product Views.
- Automated checks of parcel data processing steps.
- Automated data quality checks for future major data migrations/updates/fixes.

2.4. Standard Address Products

The table below shows the standard address products created on a regular basis:

Product	Description
Address Count Listing Files	Address Count Listing Files (count of housing units and group quarters by block) are created twice a year. Address Count Listing Files include all 50 states, DC, Puerto Rico, and the Island Areas, and they get posted on the Census website in text file format for the public to download. These count lists support our government partners and the public in tracking change in housing throughout the decade and can be useful in determining potential impact during natural disasters.

Product	Description
Economic Census Geocoding System and Reference Files	Economic Census ZIP Codes Reference File (ZIPREF). This product is delivered to EMD.
Master Address File Extracts (MAFX), and MAF Operation Codes (MAFOP), MAFX Tally Files	The MAF extracts are delivered to various customers twice a year (by March 1st and September 1st). The MAF extracts are delivered in pipe-delimited format, while the Tally files are delivered in text format. A MAF extract in Oracle Dump file format is also created and delivered to ADSD for the Listing and Mapping Application (LiMA). The MAF extracts contain updated address, geocode, source, and history data for all current counties and county equivalents in the United States, DC, and Puerto Rico and are delivered to support program in data sampling, data collection, tabulation processes and research.

2.5. Core Update Applications

The Core Update area provides centralized low-level software to support updates to the MAF/TIGER System and keep it consistent. Software includes:

- Input/Output API library that offers an interface into the Oracle Topology Data Model and the data in the MTDB to MAF/TIGER System's software developers.
- Business rules engine to maintain the integrity of the data in the MTDB.
- Legal Values Database system to maintain MTDB legal values and ensure data integrity.
- Currency Tracking System to track updates and maintain up to date on a delayed basis certain spatial attribution for features.
- Provides web based interactive MTDB data visualization tools.

2.6. Spatial and Geographic Tools and Technology

Product	Description
Geographic Metadata Repository (GMDR)	The Geographic Metadata Repository (GMDR) supports the dissemination of data within the CEDSCI system. The GMDR is made up of a series of tables that provide the following functionality within CEDSCI: Construction of a hierarchical text tree of geographic names for browsing, text search on geographic names, and the linking of summary tables with the geospatial/vector tiles API via the GEOID in order to support mapping.

Product	Description
Geographic Information Table (GIT)	The Geographic Information Table (GIT) provides an inventory of all GEOIDs for a given product. Each record also breaks out the components of the GEOID into separate fields. This is used for CEDSCI.
CEDSCI Geodatabase	The CEDSCI Geodatabase is an extract of MAFTIGER by summary level and year providing spatial data for CEDSCI. This database is used to generate vector tiles. It is also used directly by CEDSCI to perform spatial and attribute queries for selecting, highlighting, and zooming the Enterprise Data Dissemination Engine (EDDE) map component as a user interacts with the map.
CEDSCI Vector Tiles	The CEDSCI Vector Tiles support the mapping components of the EDDE. Vector tiles are packages of geographic data that contain information to draw geometries in a web map in an efficient manner. GEO/DITD create a vector tile package for each summary level needed for a particular data release. Vector tiles support the annual American Community Survey, the 2020 Census, Economic Census, and the Small Area Income and Poverty Estimates. In addition, GEO/DITD provide base maps as vector tile packages each year.
LiMA spatial extract	The LiMA spatial extracts are Oracle Spatial files that include code identifiers, attributes, and the spatial extent of physical features (e.g., roads, railroads, hydrography, and landmarks) and geographic areas, (e.g., counties, county subdivisions, places, American Indian areas). The Spatial files are delivered twice a year (January and August), following each biannual MAF/TIGER benchmark. We deliver recurring production and test versions of this intercensal products from the ACS and BAS Benchmarks.
Full spatial extract	The full spatial extract (FSE) is a file geodatabase (GDB) spatial extract of the TIGERweb GDB. The FSE includes tabulation geographic areas, collection geographic areas, and features such as roads and hydrography. The FSE is used by MAF/TIGER's Geospatial Services to support map services in the Enterprise Data Lake (EDL) cloud environment.
TIGER/Line Shapefiles	The TIGER/Line Shapefiles are extracts of selected geographic and cartographic information from the USCB's MAF/TIGER Database. The shapefiles include information for the fifty states, the District of Columbia, Puerto Rico, and the Island areas (American Samoa, the Commonwealth of the Northern Mariana Islands, Guam, and the United States Virgin Islands). The shapefiles include polygon boundaries of geographic areas and features, linear features including roads and hydrography, and point features. These shapefiles do not contain any sensitive data or confidential data protected by Title 13 of the U.S.C, and the core TIGER/Line Files and Shapefile do not include demographic data, but they do contain geographic entity codes (GEOIDs) that can be linked to the USCB's demographic data. This product is produced during the ACS benchmark and occasionally for decennial and post decennial programs (e.g., CD and SLD TIGER/Line creation).

Product	Description
TIGER/Line GDBs	TIGER Geodatabases are spatial extracts from USCB's MAF/TIGER database for use with Geographic Information Systems (GIS) software. The geodatabases contain national coverage (for geographic boundaries or features) or state coverage (boundaries within state). This product is produced during the ACS benchmark and occasionally for decennial and post decennial programs (e.g., CD and SLD TIGER/Line creation).
TIGER/Line GeoPackage Databases	TIGER/Line GeoPackage Databases are open-format spatial extracts from USCB's MAF/TIGER database for use with geographic information systems (GIS) software. These GeoPackages contain national coverage (for geographic boundaries or features) or state coverage (boundaries within state). This product is produced during the ACS benchmark and occasionally for decennial and post decennial programs (e.g., CD and SLD TIGER/Line creation).
TIGER/Line GeoPackage Files	The TIGER/Line GeoPackage Files are extracts of selected geographic and cartographic information from the USCB's MAF/TIGER Database. The GeoPackages include information for the fifty states, the District of Columbia, Puerto Rico, and the Island areas (American Samoa, the Commonwealth of the Northern Mariana Islands, Guam, and the United States Virgin Islands). The GeoPackages include polygon boundaries of geographic areas and features, linear features including roads and hydrography, and point features. These GeoPackages do not contain any sensitive data or confidential data protected by Title 13 of the U.S.C, and the core TIGER/Line GeoPackage Files do not include demographic data, but they do contain fully qualified geographic entity codes (GEOIDFQs) that can be linked to data tables on data.census.gov. This product is produced during the ACS benchmark and occasionally for decennial and post decennial programs (e.g., CD and SLD TIGER/Line creation).
Gazetteer files	The U.S. Gazetteer Files provide a listing of all geographic areas for selected geographic area types. The files include geographic identifier codes, names, area measurements, and representative latitude and longitude coordinates. Gazetteer files is an annual public product produced during the ACS benchmark.
GURL	Geographic Update Reporting Log (GURL) is an annual product provided to the ACSO from each ACS benchmark starting with a prototype delivery in 2023 and with production versions from 2024 onward. The GURL is a single replacement product superseding the GCCF and GChnCF products. The GURL provides a standardized listing of changes enacted on various tabulation geographies from one year to the next (from the previous ACS vintage to the current ACS vintage) so the ACS can account for the changes as they process and tabulate summaries.

Product	Description
Grids	This new product is defined by applying demographic, housing, and geographic variables onto a series of grid cells that are of uniform in size and shape that remains stable over time. This annual product is published for both internal and external use as well as for international geographic integration. Research is required to determine the suite of products at various grid sizes to produce.
QuickFacts	QuickFacts are provided on an annual basis and provide basic information about Census places including their population estimates either from the Decennial Census or from the most recent Population Estimates Program results, whichever is more current. They are created and delivered as .xlsx tables to the Center of New Media and Promotion (CNMP). CNMP uses the tables to update their QuickFacts website for public consumption of data.
TIGERweb Geodatabase	SDE-enabled Oracle tables extracted from a MAFTIGER benchmark, used as the source of the TIGERweb application and map services. Allows users to view and query census geographic areas and features such as roads, railroads, rivers, lakes and larger bodies of water. It currently stores boundaries, names and codes for current and 2020 Census legal and statistical geographic areas, such as counties, cities, towns, and townships, and census tracts, and urban areas. In addition, TIGERweb contains population and housing unit counts from the 2020 Census for each of the geographic areas.
TIGERweb 508 HTML files	The Nation-Based, State-Based, and County-Based tabular HTML files contain the same attribute data found in the TIGERweb Map Services. These files are provided in compliance with the 1998 Section 508 amendment of the Workforce Rehabilitation Act of the 1973.
Geographic Reference Files Codes (GRF-C)	Documents that list codes for geographic entities. These include Relationship Files, Block Assignment Files, and Tally files. Or A USCB listing all geographic codes associated with census block records. GRF products get delivered from the ACS Benchmarks to these customers: ACS, DSMD, SEHSD, POP, IRE, ERD/CODS, DSSD, LEHD, PEARSIS, PEC-PCS (2021 only), and census.gov.
Geographic Reference Files Names (GRF-N)	A USCB computer file listing the name of each geographic entity and their associated attributes. GRF products get delivered from the ACS Benchmarks to these customers: ACS, DSMD, SEHSD, POP, IRE, ERD/CODS, DSSD, LEHD, PEARSIS, PEC-PCS (2021 only), and census.gov.

Product	Description
Partnership Shapefiles	The Partnership Shapefiles are used in our partner programs to share data with and capture data from our partners. These shapefiles are intended for geographic partnership program uses only. Partnership shapefiles are produced for our local geographic partners for programs like the Boundary Annexation Survey (BAS) or School District Review Program (SDRP). We post them on our website for the Partnership participants to download and we can send them by mail on DVDs if needed. They are produced twice a year as part of GEO's ACS and BAS benchmarks. They can be produced from live MAFTIGER/PRODTRAN using LMPRE as well.
LISRDS	The Live Individual Shapefile Request and Delivery System (LISRDS) allows users to request spatial files and/or dBase III table extracts (layers) from the tables in the MAFTIGER Database (MTDB). Data can be extracted from either Live MTDB or one of the MTDB benchmarks.
Cartographic Boundary Files	Cartographic Boundary Files include generalized representations of geographic area boundaries for use in small scale thematic mapping and web mapping. They are available as shapefiles, file geodatabases, and KML files. These files are posted to census.gov and are used internally.
SDRP Detailed and Generalized GDBs	The SDRP Detailed and Generalized Geodatabase is an extract of MAFTIGER by summary level. The SEHDS-SAEB uses detailed and generalized layers for data review, analysis, and dissemination. These are produced during the ACS benchmark to support a series of SDRP product releases.
National SDRP Relationship Listings	The National SDRP relationship listings include several geographic relationship and inventory files created from MTDB. The SEHSD-SAEB use these files for estimates production and quality assurance (QA). These are produced during the ACS benchmark to support a series of SDRP product releases.
State SDRP Relationship Listings	The State SDRP relationship listings include several geographic relationship and inventory files created from the MTDB. SDRP partners use these listings during the annotation and verification phases. These are produced during the ACS benchmark to support the next SDRP cycle.
Economic Census ZIP to State/County File	The ZIPREF is used to store the geographic metadata and data needed by the edit, review, correction, tabulation, and dissemination systems for the ZIP Code-based data products from the 20xx Economic Census. This file is also used by the same systems for the related programs from Economy-Wide Statistics Division (County Business Patterns (CBP)). This product delivers to EMD.
Economic Census Codes and Names and Changes Files	This file denotes the updated EC Places that have changed codes and names. This product delivers to EMD.

Product	Description
Economic Census Qualifying Economic Place File	This file lists the current qualifying EC places for the current EC session. This product is delivered to EMD.
Economic Census Standard 'X2 to Standard 'X7 Recode File	The Economic Census Place Recode file is to aid the Economy-Wide Statistics Division (EWD) census subject area staff in verifying economic places that changed between 20xx (Previous) and 20xx (Current) as part of the 20xx(current) Economic Census analysis. This product delivers to EMD.
Economic Census MTDB from Vintage X to GCAT File	This file is to load the 20xx Economic Census data into the GCAT database on PRODTRAN. The software will utilize the attached 20xx Economic Census data migration crosswalk (GCAT_20xx_Econ_Data_Migration.xlsx) for use in migrating data from vintage 'xx' MTDB into the vintage 'xx' GCAT database.
Economic Census Geographic Area Reference File	The GARF is used to store the geographic metadata and data needed by the edit, review, correction, tabulation, and dissemination systems for the data products from the 20xx Economic Census. This file is also used by the same systems for the related programs from Economy-Wide Division (the Survey of Business Owners and the Economic Census of Island Areas), from Economic Management Division (Nonemployer Statistics and County Business Patterns (CBP)), Economy-Wide Division (Annual Survey of Manufactures), and from Economic Reimbursable Division (the Commodity Flow Survey). These reports exclude the ZIP Code data files; these codes will be stored in a separate ZIP Code Reference File specified in its own separate CRD. This product is delivered to EMD.
Economic Census Geographic User Notes	The purpose of this product is to denote all geographic and statistical changes for all designated Economic Census areas from the 20xx(previous) Economic Census to the 20xx (current)Economic Census. Economic Census Geographic User Notes product delivers this to WPSB.
Economic Census Place Recode File	This product provides requirements for two files used by the 20xx (current) Economic Census Geocoding System: The Standard 20xx(previous) Geography to Standard 20xx(current) Geography Recode File, and the Standard 20xx(current) Geography to EC 20xx(current) Geography Recode File. The first file is internal to the geocoder and will convert historical 20xx(previous) codes for state, county, MCD, and place into the current 20xx(current) code set during initial geocoding. The second file will recode Standard geography to Economic Census geography. This file will also contain other geocodes and flags required by the Economic Census Geocoding System. This product delivers to ESMD.
Data Products Production (DPP)	The Data Products Production (DPP) tables are Oracle tables that include all summary level records with geographic information reflecting the contents of the decennial geoheaders. These tables are only required for decennial deliveries.

Product	Description
GPP HEO Extract	The GPP HEO Extract will be delivered twice year to the Cybersecurity and Infrastructure Security Agency. The GPP HEO file is spreadsheet that contains HEO Contact ID, Entity Type, Entity Name, Entity ID, State Abbreviation, HEO Name, and HEO Email Address.
County Adjacency File	The county adjacency file lists each county, or county equivalent, and which county, or counties, are neighboring. This file is created and posted to the public annually.

2.7. Cartographic Applications and Services

Regularly created cartographic applications and services include the following:

Product	Description
TIGERweb Application	TIGERweb includes interactive web map applications and services that allow users to visualize the geographic boundaries and features from the MAF/TIGER database. The TIGERweb is intended to meet the needs of users inside and outside the USCB for access to geospatial data contained within the TIGER geodatabase without requiring access to a GIS or downloading data.
Generalization Systems	The generalization software performs cartographic generalization on geographic boundaries to produce products such as Cartographic Boundary Files.
Web Map Applications and Services	GEO/DITD releases interactive web map applications and services as needed to support various programs and data releases. The applications use Esri Web AppBuilder, Experience Builder, ArcGIS StoryMaps, or the Esri javascript API.
Census Automated Map Production System	The Census Automated Map Production System (CAMPS) produces reference maps in a batch environment. The software utilizes Oracle parameter tables, the MAF/TIGER database, and the cartographic database to create maps.
MAF/TIGER Resources	Provide easy access to MAF/TIGER documentation, applications, statistics, and metadata which will improve subject-matter and developer staff performance.